Counterexamples to two squared-codegree Turán conjectures

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Abstract

We study two five-vertex hypergraph Turán problems posed by Balogh, Clemen, and Lidický. For $K_5^{=}$, the complete three-uniform hypergraph on five vertices with two edges intersecting in one vertex deleted, a four-part construction disproves their squared-codegree density conjecture. It gives squared-codegree density at least $\frac{391686088}{1029059101} \approx 0.3806$ and improves the ordinary Turán lower bound to $\frac{2235+76\sqrt{114}}{5082} \approx 0.5995$. For J_4, the suspension of K_4, we correct the count for the known iterated blow-up of the Fano complement and obtain squared-codegree density at least $16/57$, exceeding the proposed value $7/25$. We further prove that $16/57$ is optimal within this recursive family, allowing arbitrary part sizes and recursion depths. The proofs use explicit constructions, finite counting identities, and a nonnegative weighted-square identity.

1 Introduction

Balogh, Clemen, and Lidický [1, 2] study hypergraph Turán problems in which the sum of squared pair codegrees replaces the number of edges. Their survey proposes numerical values for several forbidden hypergraphs. We address two of its five-vertex cases: the hypergraph $K_5^{=}$ and the suspension J_4 of K_4. For $K_5^{=}$ we construct a counterexample to the squared-codegree conjecture and improve the ordinary lower bound. For J_4 we obtain a larger lower bound by correcting the count of an existing recursive construction.

For a positive integer m, write $[m] = \{1, \dots, m\}$. For a finite set X and a nonnegative integer r, let $\binom{X}{r}$ denote the family of its r-element subsets. For a finite three-uniform hypergraph H with vertex set $V(H)$ and edge set $E(H)$, write $e(H) = |E(H)|$. For distinct $u, v \in V(H)$, let $d_H(u, v)$ be the number of edges containing $\{u, v\}$. Define

$$\mathrm{co}_2(H) = \sum_{\{u,v\} \in \binom{V(H)}{2}} d_H(u, v)^2.$$

For a fixed three-uniform hypergraph F, call H F-free if it contains no subhypergraph isomorphic to F; the copy need not be induced. Let $\mathrm{ex}(n, F)$ and $\mathrm{exco}_2(n, F)$ denote the maximum values of $e(H)$ and $\mathrm{co}_2(H)$, respectively, over all F-free hypergraphs on n vertices. We use the normalizations

$$\pi(F) = \lim_{n\to\infty} \frac{\mathrm{ex}(n, F)}{\binom{n}{3}}, \quad \text{and} \quad \sigma(F) = \lim_{n\to\infty} \frac{\mathrm{exco}_2(n, F)}{\binom{n}{2}(n-2)^2}.$$

Both limits exist; see [2, Section 1]. For construction sequences whose orders tend to infinity, write $n = |V(H)|$. The normalizations $6e(H)/n^3$ and $2\,\mathrm{co}_2(H)/n^4$ give the ordinary and squared-codegree limits, respectively. We cite the published survey as [1] and its revised version as [2]. Numbered references to their survey refer throughout to [2].

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A counterexample for $K_5^{=}$. Let $K_5^{=}$ be the three-uniform hypergraph on $[5]$ with $E(K_5^{=}) = \binom{[5]}{3} \setminus \{123, 145\}$, where a string such as 123 denotes $\{1, 2, 3\}$. Thus the two missing edges intersect in one vertex. Balogh, Clemen, and Lidický [2, Conjecture 3.6] conjecture that

$$\pi(K_5^{=}) = \frac{227}{387}, \qquad \sigma(K_5^{=}) = \frac{2909}{8127}. \tag{1.1}$$

Using the three-part construction described by Balogh, Clemen, and Lidický [2, Section 3.19], a machine-generated online comment [7, comment 1] obtains the corrected ordinary lower bound

$$\pi(K_5^{=}) \geq \frac{5}{9} + \frac{1}{9} \cdot \frac{2}{7} = \frac{37}{63} > \frac{227}{387}.$$

Our first result improves this corrected ordinary lower bound of $37/63$ and disproves the squared-codegree assertion in (1.1).

Theorem 1.1. *The ordinary and squared-codegree Turán densities of $K_5^{=}$ satisfy*

$$\pi(K_5^{=}) \geq \frac{2235 + 76\sqrt{114}}{5082} > \frac{37}{63}, \tag{1.2}$$

$$\sigma(K_5^{=}) \geq \frac{391686088}{1029059101} > \frac{2909}{8127}. \tag{1.3}$$

These lower bounds are approximately 0.5995 *and* 0.3806*, respectively, and are attained as limits by two sequences in the same four-part family.*

An improved lower bound for J_4. The hypergraph J_4 is obtained by adjoining one common vertex to each edge of K_4; explicitly, $E(J_4) = \{123, 124, 125, 134, 135, 145\}$. Balogh, Clemen, and Lidický [2, Conjecture 3.4] propose

$$\sigma(J_4) = 0.28 = \frac{7}{25}. \tag{1.4}$$

Their calculation uses balanced iterations of the complement of the Fano plane, a construction already considered by Bollobás, Leader, and Malvenuto [3] for the ordinary Turán problem. Our second result improves the lower bound for $\sigma(J_4)$ by correcting the count of this recursive construction.

Theorem 1.2. *The squared-codegree Turán density of J_4 satisfies*

$$\sigma(J_4) \geq \frac{16}{57} = \frac{7}{25} + \frac{1}{1425}.$$

Moreover, balanced iterations of the complement of the Fano plane have limiting squared-codegree density $16/57$.

Theorem 1.2 refutes (1.4). We also prove that $16/57$ is optimal within the Fano-complement recursive family, even with arbitrary part sizes and recursion depths (see Lemma 3.2).

Section 2 gives the four-part construction and proves Theorem 1.1. In Section 3, we prove Theorem 1.2 and the subsequent optimality statement for the Fano-complement recursive family.

2 A counterexample for $K_5^{=}$

We first establish a sufficient condition for a four-part construction to be $K_5^{=}$-free. We then choose the internal hypergraphs and part proportions to obtain the two bounds in Theorem 1.1.

2.1 Construction and weighted counting

Write $K_4^{(3)-}$ for the complete three-uniform hypergraph on four vertices with one edge deleted. Let $\mathcal{D}$ be a loopless directed graph on $[4]$. Partition a finite vertex set into four possibly empty parts $V_1, \dots, V_4$. For each $i \in [4]$, let H_i be a three-uniform hypergraph with vertex set V_i. Include every triple meeting three distinct parts. For each arc $i \to j$ of $\mathcal{D}$, include every triple with two vertices in V_i and one in V_j. Finally, include the edges of each H_i. Denote the resulting hypergraph by $H(\mathcal{D}; H_1, \dots, H_4)$. The use of directed graphs to prescribe edges meeting exactly two parts also appears in Markström [6].

The next lemma isolates the structural condition that permits internal edges without creating the forbidden five-vertex hypergraph.

Lemma 2.1. *Suppose that every H_i is $K_4^{(3)-}$-free. If every vertex of $\mathcal{D}$ has outdegree at most one and $\mathcal{D}$ has no directed two-cycle, then $H(\mathcal{D}; H_1, \dots, H_4)$ is $K_5^{=}$-free.*

Proof. Suppose, for a contradiction, that the construction contains a copy of $K_5^{=}$ with vertex set U. Then U has at most two missing triples, and any two missing triples must intersect in exactly one vertex. Since every four vertices of $K_5^{=}$ span at least three edges and each H_i is $K_4^{(3)-}$-free, we have $|U \cap V_i| \le 3$ for every $i \in [4]$.

Suppose that $|U \cap V_i| = 3$ for some i. For any $v \in U \setminus V_i$, say $v \in V_j$, the arc $i \to j$ must be present. Otherwise the three triples consisting of v and two vertices of $U \cap V_i$ are missing. As i has outdegree at most one, both vertices of $U \setminus V_i$ lie in the same part V_j. The absence of directed two-cycles implies that $j \to i$ is absent. These two vertices, together with each vertex of $U \cap V_i$, give three missing triples, a contradiction. Thus $|U \cap V_i| \le 2$ for every $i \in [4]$.

Since $|U| = 5$ and there are only four parts, some part satisfies $U \cap V_i = \{a, b\}$. Among the three triples $\{a, b, x\}$ with $x \in U \setminus V_i$, at most one can be missing, since any two intersect in the pair $\{a, b\}$. Thus there are distinct $c, d \in U \setminus V_i$ such that both $\{a, b, c\}$ and $\{a, b, d\}$ are edges. By the construction, the parts containing c and d must have labels that are outneighbors of i. Since i has outdegree at most one, c and d lie in the same part V_j, with $i \to j$. The absence of directed two-cycles implies that $j \to i$ is absent. Hence $\{a, c, d\}$ and $\{b, c, d\}$ are missing and intersect in the pair $\{c, d\}$, a contradiction. Therefore the construction is $K_5^{=}$-free. □

We now specify the internal hypergraphs using a classical six-vertex construction. To make the counting self-contained, define S on $[6]$ by

$$E(S) = \{123, 125, 136, 145, 146, 234, 246, 256, 345, 356\}.$$

This is the hypergraph used in the iterated construction of Frankl and Füredi [4]; see also [2, Section 3.9]. The edge list shows that every four vertices of S span exactly two edges and every pair has codegree two.

We define a sequence $(S_k)_{k \ge 0}$ recursively. Let S_0 consist of one vertex and no edges. For each $k \ge 1$, having defined S_{k-1}, take six vertex-disjoint copies of S_{k-1} with vertex sets $W_1, \dots, W_6$. To form S_k, retain all edges within these copies and, for each $\{i, j, \ell\} \in E(S)$, add every triple with one vertex in each of W_i, W_j, W_ℓ. No other edges are added. In particular, no edge meets exactly two of the sets W_i. Thus $S_1 \cong S$ and $|V(S_k)| = 6|V(S_{k-1})| = 6^k$ for $k \ge 1$.

Lemma 2.2. *For every integer $k \ge 0$, the hypergraph S_k is $K_4^{(3)-}$-free and, with $n = 6^k$ vertices, satisfies*

$$e(S_k) = \frac{n^3 - n}{21}, \quad \text{and} \quad \mathrm{co}_2(S_k) = \frac{2(n^4 - n)}{43}. \tag{2.1}$$

As $k \to \infty$, the ordinary and squared-codegree densities of S_k tend to $\rho = 2/7$ and $\tau = 4/43$, respectively.

Proof. We proceed by induction on k. The case $k = 0$ is immediate. Suppose that $k \geq 1$ and the assertions hold for S_{k-1}. Let U be a four-element subset of $V(S_k)$. If $U \subseteq W_i$ for some i, it spans at most two edges by induction. If the vertices of U lie in four distinct parts, they span exactly two edges by the four-vertex property of S. Otherwise, some part satisfies $r = |U \cap W_i| \in \{2, 3\}$. Every triple consisting of two vertices of $U \cap W_i$ and one vertex of $U \setminus W_i$ is missing. There are $\binom{r}{2}(4-r) \geq 2$ such triples, so U spans at most two edges. Hence S_k is $K_4^{(3)-}$-free.

For the counts, put $m = 6^{k-1}$, so $|W_i| = m$ and $n = 6m$. The ten edges of S contribute $10m^3$ edges meeting three parts. Since every pair of vertices of S has codegree two, each pair in distinct parts of S_k has codegree $2m$. There are $15m^2$ such pairs, contributing $15m^2(2m)^2 = 60m^4$ to $\mathrm{co}_2(S_k)$. Pairs within a part have only internal common neighbors. Consequently,

$$e(S_k) = 10m^3 + 6e(S_{k-1}), \quad \text{and} \quad \mathrm{co}_2(S_k) = 60m^4 + 6\,\mathrm{co}_2(S_{k-1}).$$

Applying the induction hypotheses to S_{k-1}, which has m vertices, and using $n = 6m$, we obtain

$$e(S_k) = 10m^3 + \frac{6(m^3 - m)}{21} = \frac{(6m)^3 - 6m}{21} = \frac{n^3 - n}{21},$$

and

$$\mathrm{co}_2(S_k) = 60m^4 + \frac{12(m^4 - m)}{43} = \frac{2((6m)^4 - 6m)}{43} = \frac{2(n^4 - n)}{43}.$$

This proves (2.1) and completes the induction. Multiplying the first by $6/n^3$ and the second by $2/n^4$ gives the limits $2/7$ and $4/43$. □

We now count hypergraphs of the form $H(\mathcal{D}; H_1, \ldots, H_4)$. Fix a loopless directed graph $\mathcal{D}$ on $[4]$ with outdegree at most one and no directed two-cycle. Let $x = (x_1, \ldots, x_4)$ be a vector of nonnegative target part proportions with $\sum_i x_i = 1$. All class indices in the following definitions and weighted formulas range over $[4]$. Define

$$L_i = \sum_{j : i \to j} x_j, \qquad C_{ij} = 1 - x_i - x_j + \mathbf{1}_{i \to j} x_i + \mathbf{1}_{j \to i} x_j \quad (i \neq j).$$

Here $\mathbf{1}_{i \to j}$ is 1 if $i \to j$ is an arc of $\mathcal{D}$ and 0 otherwise. Sums indexed by $i \to j$ run over the arcs of $\mathcal{D}$. For sequences in this family with orders tending to infinity and part proportions tending to x, L_i is the limiting proportion of common neighbors outside V_i for a pair within V_i, while C_{ij} is the limiting proportion of all common neighbors for a pair in distinct parts V_i, V_j. Both proportions are taken relative to the total number of vertices.

Lemma 2.3. *There is a sequence of $K_5^{=}$-free hypergraphs of the form $H(\mathcal{D}; H_1, \ldots, H_4)$ whose orders tend to infinity and whose part proportions tend to x. Their ordinary and squared-codegree densities converge, respectively, to*

$$P_{\mathcal{D}}(x) = 6 \sum_{i<j<\ell} x_i x_j x_\ell + 3 \sum_{i \to j} x_i^2 x_j + \rho \sum_i x_i^3, \tag{2.2}$$

$$Q_{\mathcal{D}}(x) = 2 \sum_{i<j} x_i x_j C_{ij}^2 + \sum_i x_i^2 (L_i^2 + 2\rho x_i L_i + \tau x_i^2), \tag{2.3}$$

where $\rho = 2/7$ and $\tau = 4/43$.

Proof. We form $H(\mathcal{D}; H_1, \dots, H_4)$ by choosing each internal hypergraph H_i as a suitable blow-up of S_k from Lemma 2.2.

First suppose that $x_1, \dots, x_4$ are rational. For each integer $k \geq 0$, let n be 6^k times a fixed positive common denominator of these four numbers. Thus $n \to \infty$ as $k \to \infty$. For each $i \in [4]$, replace each vertex of a separate copy of S_k by $nx_i/6^k$ independent clones, including all triples whose original vertices form an edge. Denote the resulting internal hypergraph by H_i and its vertex set by V_i. Since S_k has 6^k vertices, $|V_i| = nx_i$, with V_i empty when $x_i = 0$.

Four clones of distinct original vertices span at most two edges by Lemma 2.2. If two of the four clones have the same original vertex, both triples containing that pair are absent, so again at most two edges remain. Thus each H_i is $K_4^{(3)-}$-free, and Lemma 2.1 shows that $H = H(\mathcal{D}; H_1, \dots, H_4)$ is $K_5^{=}$-free.

Cloning multiplies the edge count by the cube of the number of clones per vertex, and the squared-codegree sum by its fourth power since a pair of distinct original vertices gives the square of that number of clone pairs, each with internal codegree multiplied by the same number. Two clones of the same vertex have codegree zero in H_i. Consequently, (2.1) gives

$$\frac{6e(H_i)}{n^3} = \frac{6x_i^3}{6^{3k}} e(S_k) = \rho x_i^3 (1 - 6^{-2k}), \quad \text{and} \quad \frac{2\,\mathrm{co}_2(H_i)}{n^4} = \frac{2x_i^4}{6^{4k}} \mathrm{co}_2(S_k) = \tau x_i^4 (1 - 6^{-3k}).$$

Counting edges that meet three parts, two parts, or one part, and using $|V_i| = nx_i$, we obtain the exact identity

$$\frac{6e(H)}{n^3} = 6 \sum_{i<j<\ell} x_i x_j x_\ell + 3 \sum_{i \to j} x_i \left(x_i - \frac{1}{n} \right) x_j + \sum_i \frac{6e(H_i)}{n^3}.$$

Substituting the first internal count above and letting $k \to \infty$ gives (2.2).

To count squared codegrees, let $\varepsilon_{ij} = \mathbf{1}_{i \to j} + \mathbf{1}_{j \to i}$. For $u \in V_i$ and $v \in V_j$ with $i \neq j$, the codegree is

$$d_H(u, v) = n(1 - x_i - x_j) + \mathbf{1}_{i \to j}(nx_i - 1) + \mathbf{1}_{j \to i}(nx_j - 1) = nC_{ij} - \varepsilon_{ij}.$$

The terms $nx_i - 1$ and $nx_j - 1$ exclude the endpoints u and v, respectively. For distinct $u, v \in V_i$, we have $d_H(u, v) = nL_i + d_{H_i}(u, v)$. Expanding the square gives

$$d_H(u, v)^2 = n^2 L_i^2 + 2nL_i\, d_{H_i}(u, v) + d_{H_i}(u, v)^2.$$

When we sum over all pairs in V_i, the internal codegrees sum to $3e(H_i)$, because each internal edge contains three pairs. Hence the middle terms contribute $2nL_i \cdot 3e(H_i) = 6nL_i e(H_i)$, while the last terms contribute $\mathrm{co}_2(H_i)$. There are $n^2 x_i x_j$ pairs between V_i and V_j, and $\binom{nx_i}{2}$ pairs within V_i. Summing their squared codegrees and multiplying by $2/n^4$ gives

$$\frac{2\,\mathrm{co}_2(H)}{n^4} = 2 \sum_{i<j} x_i x_j \left(C_{ij} - \frac{\varepsilon_{ij}}{n} \right)^2 + \sum_i \left[x_i \left(x_i - \frac{1}{n} \right) L_i^2 + 2L_i \frac{6e(H_i)}{n^3} + \frac{2\,\mathrm{co}_2(H_i)}{n^4} \right].$$

Substituting the two internal counts above and letting $k \to \infty$, the three terms in brackets tend to $x_i^2 L_i^2$, $2\rho x_i^3 L_i$, and τx_i^4, respectively. The first sum tends to $2\sum_{i<j} x_i x_j C_{ij}^2$, proving (2.3).

Finally, we extend the construction to real part proportions; this allows us to use the irrational proportions in the proof of Theorem 1.1. Given any such x, choose a sequence $(x^{(j)})_{j \geq 1}$ of rational four-dimensional vectors with nonnegative coordinates summing to one and $x^{(j)} \to x$. For each integer $j \geq 1$, the rational case provides an $K_5^{=}$-free hypergraph H on $n \geq j$ vertices with part proportions $x^{(j)}$, such that its two normalized densities differ from $P_{\mathcal{D}}(x^{(j)})$ and $Q_{\mathcal{D}}(x^{(j)})$, respectively, by less than $1/j$. For these chosen graphs, $\left| \frac{6e(H)}{n^3} - P_{\mathcal{D}}(x) \right| \leq \frac{1}{j} + \left| P_{\mathcal{D}}(x^{(j)}) - P_{\mathcal{D}}(x) \right|$, and $\left| \frac{2\,\mathrm{co}_2(H)}{n^4} - Q_{\mathcal{D}}(x) \right| \leq \frac{1}{j} + \left| Q_{\mathcal{D}}(x^{(j)}) - Q_{\mathcal{D}}(x) \right|$. Both right-hand sides tend to zero by continuity of the two polynomials. Since $n \geq j$ and $x^{(j)} \to x$, these graphs also have orders tending to infinity and part proportions tending to x, as required. □

2.2 The two density bounds

Proof of Theorem 1.1. We use the same four-part family for both bounds, with different part proportions. Relabel the four parts $1, 2, 3, 4$ as A, B, C, D, respectively, and write $x = (x_A, x_B, x_C, x_D)$. Let $\mathcal{D}$ have the arcs

$$A \to B, \quad B \to C, \quad C \to A, \quad D \to A.$$

Every vertex of $\mathcal{D}$ has outdegree one, and there is no directed two-cycle. Thus Lemma 2.3 provides an $K_5^=$-free sequence with limiting densities $P_{\mathcal{D}}(x)$ and $Q_{\mathcal{D}}(x)$ for every vector x of nonnegative part proportions summing to one. In particular, $\pi(K_5^=) \geq P_{\mathcal{D}}(x)$ and $\sigma(K_5^=) \geq Q_{\mathcal{D}}(x)$. It remains to choose suitable proportions for each bound.

For the ordinary density, let $t \in [0, 1]$ be a parameter to be chosen later, and set $x_A = x_B = x_C = (1 - t)/3$ and $x_D = t$. Writing $p(t) = P_{\mathcal{D}}(x)$, substitution in (2.2) gives

$$p(t) = \frac{37}{63}(1 - t)^3 + 2t(1 - t)^2 + t^2(1 - t) + \frac{2}{7}t^3 = \frac{37 + 15t - 78t^2 + 44t^3}{63}.$$

Its derivative is $(15 - 156t + 132t^2)/63$. The two roots are $(13 \pm \sqrt{114})/22$, and only the smaller root lies in $[0, 1]$. On $[0, 1]$, the derivative is positive before that root and negative after it. Therefore

$$\max_{0 \leq t \leq 1} p(t) = p\left(\frac{13 - \sqrt{114}}{22}\right) = \frac{2235 + 76\sqrt{114}}{5082}. \tag{2.4}$$

This proves (1.2). Since $p(0) = 37/63$ and $p'(0) > 0$, the bound strictly exceeds the corrected three-part value. The maximum in (2.4) concerns only the indicated one-parameter subfamily.

For the squared-codegree density, take $x = \left(\frac{13}{43}, \frac{12}{43}, \frac{12}{43}, \frac{6}{43}\right)$ in the order A, B, C, D. Then $(L_A, L_B, L_C, L_D) = \left(\frac{12}{43}, \frac{12}{43}, \frac{13}{43}, \frac{13}{43}\right)$, and

$$(C_{AB}, C_{AC}, C_{AD}, C_{BC}, C_{BD}, C_{CD}) = \left(\frac{31}{43}, \frac{30}{43}, \frac{30}{43}, \frac{31}{43}, \frac{25}{43}, \frac{25}{43}\right).$$

Substitution in (2.3) gives

$$Q_{\mathcal{D}}(x) = \frac{1}{43^4}\left(1253292 + \frac{4 \cdot 72372}{7} + \frac{4 \cdot 71329}{43}\right) = \frac{391686088}{1029059101} = \frac{2909}{8127} + \frac{90033181}{3969227961}.$$

Together with Lemma 2.3, this proves (1.3) and completes the proof. □

The two lower bounds use different proportions. Neither the full four-variable optimization nor the exact values of $\pi(K_5^=)$ and $\sigma(K_5^=)$ are determined here.

3 An improved lower bound for J_4

The link of a vertex v in a three-uniform hypergraph is the graph on the remaining vertices in which uw is an edge exactly when vuw is a hyperedge. A hypergraph is J_4-free precisely when all its vertex links are K_4-free. We use this characterization to verify the Fano construction and then count its squared codegrees.

Proof of Theorem 1.2. Let B be the complement of a Fano plane on vertex set $[7]$. The triples of the plane are called its lines; these are precisely the triples missing from B. The hypergraph B has 28 edges; each pair has codegree four. At a fixed vertex, its link is K_6 minus a perfect matching, that is, $K_{2,2,2}$. Set B_0 to be a single vertex with no edges. For

each integer $k \geq 1$, construct B_k from seven vertex-disjoint copies of B_{k-1} labelled by the vertices of B. Retain the edges within each copy, and add every triple meeting three distinct copies whose labels form an edge of B. No triples meeting exactly two copies are added. In the link of a vertex of B_k, the portion outside its first-level part is a blow-up of $K_{2,2,2}$, hence a complete tripartite graph. The portion inside that part is a vertex link in B_{k-1}, and there are no link edges between these two portions. Induction shows that the entire link has no K_4. Thus B_k is J_4-free.

For $k \geq 1$, put $n = 7^k$ and $m = 7^{k-1}$. Pairs in distinct first-level parts have codegree $4m$. There are $21m^2$ such pairs, contributing $21m^2(4m)^2 = 336m^4$ to $\mathrm{co}_2(B_k)$. Pairs within one part have no common neighbors outside that part, since no edge meets exactly two parts. Thus each of the seven parts contributes precisely $\mathrm{co}_2(B_{k-1})$, and

$$\mathrm{co}_2(B_k) = 336 \cdot 7^{4(k-1)} + 7\,\mathrm{co}_2(B_{k-1}).$$

Dividing the recurrence by 7^{4k} gives a factor 7^{-3} in front of the preceding normalized term. Starting from $\mathrm{co}_2(B_0) = 0$, we obtain

$$\frac{\mathrm{co}_2(B_k)}{7^{4k}} = \frac{336}{7^4} \sum_{j=0}^{k-1} 7^{-3j} = \frac{8}{57}(1 - 7^{-3k}),$$

where the last equality uses the geometric sum and $336/[7^4(1 - 7^{-3})] = 8/57$. Since $n = 7^k$, this gives the exact count

$$|V(B_k)| = n = 7^k, \quad \text{and} \quad \mathrm{co}_2(B_k) = \frac{8(n^4 - n)}{57}. \tag{3.1}$$

Consequently,

$$\frac{2\,\mathrm{co}_2(B_k)}{n^4} = \frac{16}{57}(1 - n^{-3}) \longrightarrow \frac{16}{57}.$$

Since every B_k is J_4-free, this limit proves the asserted lower bound. □

The factor seven in the recurrence accounts for all seven recursive parts. After normalization by n^4, successive levels therefore contribute in the ratio $7/7^4 = 1/343$. The calculation in [2, Section 3.15] uses $1/7^4$ instead. Correcting this factor gives $16/57$ and refutes the numerical equality $\sigma(J_4) = 0.28$ in Conjecture 3.4 of that version, using the same underlying construction.

Optimality within the recursive family. We next show that $16/57$ remains optimal when class sizes and recursion depths vary. Let B again denote the Fano complement. A *Fano-complement recursive construction* is specified by a finite rooted tree with independent vertex classes at its leaves. Each internal node partitions its vertex set into at most seven children labelled by distinct vertices of B. Include all triples meeting three children whose labels form an edge of B, and no triples meeting exactly two children. Edges within each child are defined recursively. This is a special case of the recursive patterns of Liu and Pikhurko [5, Definition 1.1].

For a real weight vector $x = (x_1, \ldots, x_7)$ on the vertices of B, put $W = \sum_{i=1}^{7} x_i$, $d_{ij} = \sum_{k:\, ijk \in E(B)} x_k$ for distinct $i, j \in [7]$, and $N(x) = 2\sum_{i<j} x_i x_j d_{ij}^2$. The weights need not sum to one. When the weights are class sizes, d_{ij} is the codegree of a pair between classes i, j, and $N(x)/2$ is the total squared-codegree contribution from pairs in distinct classes. The following lemma bounds this contribution.

Lemma 3.1. *For nonnegative weights,* $N(x) \leq \frac{16}{57}\left(W^4 - \sum_i x_i^4\right)$.

Proof. For each $1 \le i < j \le 7$, let k be the third vertex on their Fano line. Label the remaining vertices a, b, c, d so that kab and kcd are lines. Then $d_{ij} = x_a + x_b + x_c + x_d$. For all real weights, $16\left(W^4 - \sum_i x_i^4\right) - 57N(x)$ equals

$$\sum_{i<j}(x_i - x_j)^2\Big[12d_{ij}^2 + 2(8x_i + 8x_j + x_k)d_{ij} + 24(x_a + x_b)(x_c + x_d) + 58(x_ax_b + x_cx_d)\Big].$$

To check the identity, expand the bracket for each pair i, j. For $u \in \{a, b, c, d\}$, the coefficients of $x_u^2, x_ix_u, x_jx_u, x_kx_u$ are $12, 16, 16, 2$, respectively. The coefficients of x_ax_b and x_cx_d are $24 + 58 = 82$; the other mixed products among x_a, x_b, x_c, x_d have coefficient $24 + 24 = 48$. Multiplication by $x_i^2 + x_j^2 - 2x_ix_j$ and summation over pairs give the right-hand coefficients in the table below, where $p, q, r, s \in [7]$ are distinct.

Monomial	Fano condition	Left-hand side	Right-hand side
x_p^4	—	0	0
$x_p^3x_q$	—	64	$4 \cdot 16 = 64$
$x_p^2x_q^2$	—	96	$2 \cdot 4 \cdot 12 = 96$
$x_p^2x_qx_r$	pqr is a line	192	$4 \cdot 48 = 192$
$x_p^2x_qx_r$	pqr is not a line	78	78
$x_px_qx_rx_s$	contains a line	-300	-300
$x_px_qx_rx_s$	contains no line	-984	-984

If pqr is not a Fano line, the x_p^2 terms from the six summands indexed by pairs containing p contribute $2 \cdot 16 + 2 \cdot 2 + 48 + 82$ to the coefficient of $x_p^2x_qx_r$. The mixed terms from the pairs $\{p, q\}$, $\{p, r\}$, and $\{q, r\}$ contribute $-2(16 + 16 + 12)$. Their sum is 78, matching $16 \cdot 12 - 57 \cdot 2$ on the left. For a monomial with four distinct indices, only mixed terms contribute on the right: its coefficient is $-2(3 \cdot 2 + 3 \cdot 48) = -300$ if the four points contain a Fano line, and $-2 \cdot 6 \cdot 82 = -984$ otherwise. On the left, the corresponding coefficients of $N(x)$ are $4 \cdot 3 = 12$ and $4 \cdot 6 = 24$, giving $16 \cdot 24 - 57 \cdot 12 = -300$ and $16 \cdot 24 - 57 \cdot 24 = -984$. The table covers all monomials of degree four, proving the identity. For nonnegative weights, its right-hand side is nonnegative, and the inequality follows. □

Lemma 3.2. *Every Fano-complement recursive construction H on $n \ge 1$ vertices is J_4-free and satisfies*

$$\mathrm{co}_2(H) \le \frac{8}{57}(n^4 - n).$$

The balanced iterations on 7^k vertices attain equality for every integer $k \ge 0$. Consequently, the supremum of the limiting squared-codegree densities in this family is $16/57$, even when class sizes and recursion depths are chosen independently at different nodes.

Proof. We induct on the height of the recursion tree. If the root is a leaf, then H is independent, so it is J_4-free and $\mathrm{co}_2(H) = 0 \le 8(n^4 - n)/57$. Otherwise, let $H_1, \dots, H_7$ be the hypergraphs in the root's children, using a hypergraph with no vertices for each unused label. Put $n_i = |V(H_i)|$, so that $n = \sum_i n_i$. As in the proof of Theorem 1.2, each vertex link is the disjoint union of its link within its child class and a complete tripartite graph, possibly with empty parts. Both are K_4-free by induction and construction, respectively, so H is J_4-free.

There are n_in_j pairs between children i, j, each with codegree $\sum_{k:\, ijk \in E(B)} n_k$. Pairs within one child have no common neighbors outside that child, since no edge meets exactly two children. Hence

$$\mathrm{co}_2(H) = \sum_i \mathrm{co}_2(H_i) + \sum_{i<j} n_in_j \left(\sum_{k:\, ijk \in E(B)} n_k\right)^2 = \sum_i \mathrm{co}_2(H_i) + \frac{1}{2}N(n_1, \dots, n_7).$$

Here $N(n_1, \dots, n_7)$ is $N(x)$ evaluated at $x_i = n_i$; its half is exactly the contribution from pairs in distinct children. Since $W = \sum_i n_i = n$, Lemma 3.1 bounds this term. The induction hypothesis bounds $\mathrm{co}_2(H_i)$ for each nonempty child; empty children contribute zero. Thus

$$\mathrm{co}_2(H) \le \frac{8}{57}\left(n^4 - \sum_i n_i^4\right) + \frac{8}{57}\sum_i (n_i^4 - n_i) = \frac{8}{57}(n^4 - n).$$

The last equality uses the cancellation of the fourth-power terms and $\sum_i n_i = n$. Hence $2\,\mathrm{co}_2(H)/n^4 \le (16/57)(1 - n^{-3})$, giving limiting squared-codegree density at most $16/57$. By (3.1), the balanced iterations attain equality in the finite bound and have densities tending to $16/57$. □

Theorem 1.2 and Lemma 3.2 support $\sigma(J_4) = 16/57$ as a natural numerical revision of the conjecture in [2, Conjecture 3.4]. This retains the proposed extremal construction with its corrected density.

Declaration of AI use

The author used GPT-6 to assist with exploring proof strategies, checking proofs, and improving the exposition. All mathematical arguments, results, and conclusions were reviewed and verified by the author, who takes full responsibility for the content of this paper.

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